Resolution Geometry: Both Sides of Linear Gravity

Hidden Values, Observers, Compatibility Selection, and the Second-Order Lock

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

October 2026

Abstract

General relativity (GR) has two sides: a source, the stress tensor, and a geometric response. Resolution Geometry (RG) now computes, at linear order, what each side *hides*, how the hidden values are recovered, and which coupling between them is compatible. It is not yet a derivation of GR, and every remaining gate is named.

Hidden and null. The Kalman decomposition of the protocol *history* separates two kinds of unseen structure. *Hidden* values are invisible now but revealed by the history: the angular moments of order $l \geq 3$ behind the stress tensor, and the two transverse-traceless (TT) values of geometry. *Null* values are never observable; they are operationally null, which is not automatically gauge.

Source side. An observer recovers hidden moments from noisy stress histories within deterministic error bounds (48 cases). Free streaming admits no finite closure. Collisions slave the hidden moments and yield a diffusion closure, asymptotically.

Geometry side. A gravitational wave has $T = 0$ but nonzero tidal curvature. A gravitational observer recovers four real coefficients per mode (48 cases).

Coupling. Conservation of the source together with RG's descent to the gauge quotient selects the linearized Einstein operator uniquely among second-order operators, verified by 80 independent holdouts.

The second-order gate. Compatibility does not fix the derivative order. At fourth order the compatible operators form a two-dimensional family, spanned by k^2E and the R-operator. They add 6 or 1 hidden values per point and give Yukawa corrections $-\frac{4}{3}e^{-r/\lambda_2}$ and $+\frac{1}{3}e^{-r/\lambda_0}$ to the Newtonian potential. Stability excludes the spin-2 ghost but not the healthy $f(R)$ scalar. Second order is therefore *selected*, by minimal hidden state (H-min, RG's analogue of Čencov's monotonicity), and confirmed by inverse-square data down to a declared scale. Below that scale higher-order completions remain a CORRESPONDENCE.

Evidence. Executed batteries: 655/655 (both-sides audit, independently verified), and the second-order battery (S1–S6, 3 canaries). Two errata to the earlier hidden-dynamics battery, an observer sign error and a flattened Hankel matrix, were found by the audit and are corrected.

Contents

1 Status first **2**

2 Hidden values versus null values **2**

3 The source side **3**

4 The geometry side **3**

5 The coupling: compatibility selection **4**

6 The second-order gate **4**

7 Main theorem **5**

8 Gate ledger and next locks **5**

9 Errata to the hidden-dynamics battery **6**

10 Batteries **6**

11 Non-claims **7**

1 Status first

Gate	Status	Where
Hidden versus null (Kalman)	locked	Thm 2.2
Source observer and task descent	locked, conditional on declared transport	Thm 3.1
Source closure	no finite closure under free streaming; asymptotic under collisions	Thm 3.2
Geometry hidden state and observer	locked (synthetic TT histories)	Thm 4.1
Linear operator	selected by conservation and descent, within second order	Thm 5.2
Derivative order	selected by H-min; data-locked above a scale	Thms 6.1–6.6
Spin-2 carrier, flat background, locality	assumed (H-carrier, H-loc)	§8
Source microdynamics	assumed (free streaming, BGK)	§8
Nonlinear completion	not derived	§8
Source–gravity backreaction	not derived	§8
Coupling scale G	unit anchor (unfixed)	§8
Full GR from RG	**not established**	—

2 Hidden values versus null values

Definition 2.1. Let a declared linear dynamics $\dot{x} = Ax$ be observed through $y = Cx$. The *instant protocol* observes $Cx(t)$ at a single time; the *history protocol* observes $Cx(\cdot)$ over an interval.

- A state direction is *hidden* if it lies in $\ker C$ but outside the unobservable subspace $\bigcap_n \ker CA^n$.
- It is *null* if it lies in the unobservable subspace.

Theorem 2.2 (Hidden versus null)**.** *Consider free streaming at nonzero wave number on 48 angular bins, observed by the* 2+1 *stress protocol.*

- *The instant protocol has rank* 5*.*
- *The history protocol has Popov–Belevitch–Hautus (PBH) rank* 48*: every hidden moment is observable.*
- *Two nonnegative states with equal stress at* $t = 0$ *differ later.*
- *A decoupled internal variable gives PBH rank* 48/49*: it is operationally null. This means unobservable* for this protocol*; it is not automatically a gauge redundancy.*
- *At* $k = 0$ *the history rank stays* 5*: hidden values need transport to become visible.*

Proof. For diagonal A, the PBH test reduces to the rank of C restricted to each eigenspace. The witnesses are in `rg_hidden_dynamics.py` (H1, H3) and in the both-sides audit (check `zero_wave_number_history_rank5`). □

3 The source side

Theorem 3.1 (Source observer and task descent)**.** *Declare a Fourier smoothness class, using 13 nonredundant real coefficients, and componentwise noise bounds. A hidden moment task is then recovered from the stress history within a deterministic bound. Across 48 cases (sine and cosine hidden moments) the maximum task error is* 3.7×10^{-8}*, below every declared bound, and the maximum future-stress holdout error is* 1.7×10^{-8}*. A misspecified smoothness class is detected through its history residual (*0.15*).*

Theorem 3.2 (Closure)**.** (i) *Under free streaming, a generic single trajectory excites* $N/2+1$ *distinct streaming frequencies (*25*,* 49*,* 97 *for* $N = 48$*,* 96*,* 192*). That is the dimension of its minimal autonomous realization, which is unbounded in the continuum. The resolved block-Hankel rank at* 10^{-12} *is* 24–25*. Hidden values are observable in principle, with exponentially degrading conditioning.*

(ii) *Isotropic BGK collisions at rate* ν *slave the hidden moments: the density follows* $D = 1/(2\nu)$ *with maximum error* 8.5×10^{-2}*,* 6.8×10^{-3} *and* 4.2×10^{-4} *for* $\nu = 2$*,* 8*,* 32*. This closure is asymptotic. BGK conserves density, not momentum, so it exchanges momentum with an implicit bath and is not a complete conserved GR source.*

4 The geometry side

Theorem 4.1 (Geometry carries its own hidden values)**.** *The plane wave* $h_{xx} = -h_{yy} = A\cos\omega(t-z)$ *has linearized Ricci tensor* 0*, so* $T = 0$ *at all times, and tidal component* $R_{txtx} = \frac{1}{2}A\omega^2\cos\omega(t-z) \neq 0$*. Source histories therefore cannot determine an added homogeneous wave. A gravitational observer recovers the four real initial coefficients of a known-frequency mode from synthetic TT histories in 48 cases, with maximum state error* 1.3×10^{-9} *and maximum held-out tidal error* 3.1×10^{-10}*. Instantaneous observations and polarization-blind detectors do not identify them.*

Remark 4.2. This sharpens the spacetime paper. The *null* part of spacetime (coordinate values, gauge) is declared. The *hidden* part, the TT data, is real: it is the state that the geometric dynamics carries.

5 The coupling: compatibility selection

Hypothesis 5.1 (H-carrier, H-loc)**.** The geometric hidden state is carried by a symmetric rank-2 field h on a flat Lorentzian background. The response operator is local, linear, Lorentz-covariant, homogeneous in derivatives, and has constant coefficients.

Theorem 5.2 (Compatibility selects the Einstein operator)**.** *Take the general second-order operator*

$$E = c_1 k^2 h + c_2(k\,k\cdot h + k\cdot h\,k) + c_3 kk\,\mathrm{tr}\,h + c_4\eta k^2\,\mathrm{tr}\,h + c_5\eta\,(k\cdot h\cdot k).$$

- *Conservation, $k\cdot E \equiv 0$, alone leaves two free coefficients. This is the source side: declared transport conserves T.*
- *Descent to the gauge quotient, $E[k\xi + \xi k] \equiv 0$, alone leaves two. This is RG's principle that dynamics must not depend on null directions.*
- *Together they leave one: $(c_i) \propto (1, -1, 1, -1, 1)$, the linearized Einstein operator (with $E = -2G_{\mathrm{lin}}$ in the derivative-symbol convention).*

The coupling scale is a unit anchor. The result is verified by exact rational constraint matrices and by 80 independent numeric holdouts against a separately assembled linearized Ricci tensor, with maximum residual 3×10^{-14}.

6 The second-order gate

Theorem 6.1 (Order is a separate gate; S1)**.** *In the general six-term fourth-order ansatz, the operators compatible with conservation and descent form a two-dimensional space,*

$$\mathrm{span}\{\, k^2 E,\ (\eta k^2 - kk)\, R_{\mathrm{lin}} \,\}, \qquad R_{\mathrm{lin}} = k\cdot h\cdot k - k^2\,\mathrm{tr}\,h.$$

Compatibility fixes the operator at *each order; it does not fix the order.*

Theorem 6.2 (Spin content and static response; S2, S3)**.** *On a spin-2 TT tensor and a spin-0 transverse-trace tensor at static k:*

- *E acts as $(k^2, -2k^2)$;*
- *$k^2 E$ acts as k^2 times that;*
- *the R-operator acts as $(0, -3k^4)$.*

For a static point source, partial fractions of the source-saturated amplitude of $E+\alpha k^2 E+\beta R$-op give

$$V(r) = -\frac{GM}{r}\left[1 - \tfrac{4}{3}e^{-r/\sqrt{\alpha}} + \tfrac{1}{3}e^{-m_0 r}\right], \qquad m_0^2 = \frac{2}{2\alpha + 3\beta},$$

which is the Stelle form [1]. Setting $\beta = -2\alpha/3$ removes the scalar, leaving the Weyl^2-type case. Only $\alpha = \beta = 0$ gives an exactly inverse-square energy channel.

Theorem 6.3 (Stability is not enough; S4)**.** *Take the exchange residues between conserved sources, normalized so that the massless graviton has $+2/3$.*

- *The massive spin-2 pole of $k^2 E$ has residue $-2/3 < 0$: it is a ghost, excluded by stability.*
- *The massive scalar of the R-operator has residue $+1/6 > 0$: it is the healthy $f(R)$ scalaron.*

Stability therefore excludes $k^2 E$ but not the R-operator. This agrees with the audit's finding that a blanket Ostrogradsky argument is insufficient.

Theorem 6.4 (Minimal hidden state selects second order; S5)**.** *Count hidden values per point: 2 per massless graviton, 5 per massive spin-2 pole, 1 per massive spin-0 pole. The compatible operators carry:*

Operator	*Hidden values per point*
E *(GR)*	*2*
$E + \beta R$*-op* $(f(R)$*-type)*	*3*
$E + \alpha k^2 E + \beta R$*-op,* $\beta = -\frac{2}{3}\alpha$ *(Weyl²-type)*	*7*
$E + \alpha k^2 E$ *(generic quadratic)*	*8*

The second-order operator is the unique compatible operator of minimal hidden state. Every higher-order completion adds hidden values that are real*, because they appear as Yukawa forces; they are not null.*

Hypothesis 6.5 (H-min)**.** Among compatible dynamics, select the minimal realization: no hidden value that the declared observable histories do not require.

H-min plays the role for the geometric order that H-mono (Čencov) plays for the Fisher metric. It is a declared selection principle, not a theorem of the equality-of-law relation.

Theorem 6.6 (Data lock down to a scale; S6)**.** *Suppose the inverse-square law holds to fractional precision ε at separations $r \geq r_t$. Then the Yukawa ranges obey*

$$\lambda_2 \leq r_t / \ln\!\left(\tfrac{4}{3}/\varepsilon\right), \qquad \lambda_0 \leq r_t / \ln\!\left(\tfrac{1}{3}/\varepsilon\right).$$

Above that scale, second order is locked by data. Below it, higher-order completions are a CORRESPONDENCE. For example, $\varepsilon = 10^{-3}$ at $r_t = 50\,\mu m$ gives $\lambda_2 \leq 6.9\,\mu m$ (illustrative values, not a cited measurement).

7 Main theorem

Main Theorem (Both-sides lock, linear order). *Assume H-carrier, H-loc and declared source transport that conserves T. Then:*

(1) *The source's hidden state is computed by history observability. Observers recover it within declared bounds, and it has no finite closure unless collisions slave it.*

(2) *The geometry's hidden state is 2 TT values per point, real and recoverable from tidal histories.*

(3) *Conservation and RG descent select the linearized Einstein operator uniquely at second order.*

(4) *H-min selects second order among all compatible orders up to four, and inverse-square data lock it above a declared scale.*

(5) *The coupling scale is a unit anchor.*

Both sides of linear *GR are thereby computed by RG modulo four declarations: H-carrier, H-loc, H-min and the source microdynamics.*

8 Gate ledger and next locks

Gate	Status	Next lock
Source moment law	assumed; observer computed	—
Source microdynamics	assumed (free streaming, BGK)	geodesic coupling from conservation plus descent
Linear operator	selected (Thm 5.2)	—
Derivative order	selected by H-min, data above scale	stronger tests push the scale down
Spin-2 carrier, flat background, locality	assumed	carrier selection among scalar, vector and tensor
Nonlinear completion	not derived	order-by-order conservation and descent (Deser bootstrap)
Source–gravity backreaction	not derived	coupled kinetic–geometric evolution
Gravitational observer on real data	blocked (no detector operator)	H1/L1 response and initial-data inversion
Coupling scale	unfixed; unit anchor	—
Full GR from RG	**not established**	all of the above

9 Errata to the hidden-dynamics battery

The both-sides audit found two errors in `rg_hidden_dynamics.py`. Both are corrected here; the original file is retained unchanged.

1. **Observer sign error.** A sine-mode hidden moment of $+0.3$ was reconstructed as -0.3. The cosine-only example had concealed it. The corrected observer uses 13 nonredundant real Fourier coefficients and is tested on both sine and cosine modes.

2. **Flattened Hankel matrix.** The Hankel matrix flattened the six output channels into one. It is replaced by a 480×80 block Hankel. The single-trajectory frequency bound is stated as such, not as a universal closure theorem.

3. **Wording.** "Null" for a decoupled variable is restated as *operationally null for the protocol*, not gauge.

10 Batteries

- `rg_both_sides_final.py`: 655/655 checks, 48 source and 48 gravitational observer cases, 80 operator holdouts. An independent verifier (QR replays and arbitrary symbolic identities) confirms it without importing the battery module.

- `rg_second_order.py`: S1–S6 by exact sympy nullspaces, spin projectors, partial fractions and residues. Three canaries are rejected: compatibility alone fixes order; stability alone fixes order; higher-order completions add no hidden state.

- `rg_hidden_dynamics.py`: original, with errata.

Limits. All data are synthetic. The analysis is linearized on a flat background. The spin counting uses the standard propagator pole rule. The inverse-square example values are illustrative. No real detector inversion is performed.

11 Non-claims

1. GR is not derived from RG.
2. Fourth-order gravity is not ruled out below the data scale.
3. No new physical prediction is made.
4. Novelty is claimed only for the RG typing (hidden versus null, compatibility selection, H-min) and its batteries; the uniqueness of linearized GR and the Stelle form are classical [1, 2, 3].

References

[1] K. S. Stelle, Classical gravity with higher derivatives, *Gen. Relativ. Gravit.* 9 (1978) 353.

[2] S. Deser, Self-interaction and gauge invariance, *Gen. Relativ. Gravit.* 1 (1970) 9.

[3] S. Weinberg, Photons and gravitons in perturbation theory: derivation of Maxwell's and Einstein's equations, *Phys. Rev.* 138 (1965) B988.

[4] R. E. Kalman, Mathematical description of linear dynamical systems, *J. SIAM Control* 1 (1963) 152.

[5] M. Rada, *Resolution Geometry: Undermining Spacetime*, 2026.

[6] M. Rada, *Resolution Geometry: The Physical Path*, 2026.